

ALEXANDRIA ESLE, LA, USA

WMO: 722487

Lat: 31.395N

Lon: 92.291W

Elev: 36

StdP: 100.89

Time Zone: -6.00 (NAC)

Period: 96-19

WBAN: 13935

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.4	-1.9	-9.7	1.6	2.2	-7.3	2.0	3.8	8.3	18.9	7.4	18.1	0.7	30	0.362

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.3	36.6	24.9	35.4	25.1	34.2	25.0	26.9	32.3	26.5	32.1	26.1	31.6	2.7	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.7	21.0	28.7	25.1	20.4	28.4	24.7	19.8	28.2	84.4	32.3	82.4	32.1	80.8	31.6	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.2	5.9	5.2		-6.6	38.1	2.0	2.0	-8.0	39.6	-9.2	40.8	-10.3	41.9	-11.8	43.4	
			DB	-7.3	27.9	1.9	0.9	-8.7	28.5	-9.9	29.0	-10.9	29.5	-12.4	30.2	
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	19.3	9.6	11.7	15.5	19.2	23.4	26.7	28.1	28.1	25.3	19.5	13.9	10.4
	DBStd	7.88	5.90	5.45	4.92	4.04	3.22	2.08	1.70	1.94	2.90	4.44	4.94	5.53
	HDD10.0	212	81	39	11	1	0	0	0	0	0	1	19	62
	HDD18.3	1073	276	195	115	38	4	0	0	0	1	41	149	253
	CDD10.0	3616	69	88	181	275	417	501	561	560	459	295	135	76
	CDD18.3	1435	6	11	27	63	163	251	302	302	210	77	15	8
	CDH23.3	14424	28	54	193	587	1590	2572	3256	3228	1994	767	117	37
CDH26.7	6654	4	4	26	136	629	1251	1669	1715	947	256	15	2	
Wind	WSAvg	1.9	2.2	2.4	2.3	2.5	2.0	1.7	1.5	1.4	1.4	1.5	1.7	2.0
Precipitation	PrecAvg	1347	131	107	131	110	130	69	128	100	97	77	100	165
	PrecMax	1931	303	201	306	226	283	179	254	194	242	246	224	332
	PrecMin	840	30	48	25	7	44	13	42	11	30	7	2	56
	PrecStd	275	72	51	81	75	65	51	56	51	62	68	58	77
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.3	27.3	28.9	31.0	34.1	37.3	37.8	38.4	36.6	33.1	28.7	26.5
		MCWB	19.0	20.0	20.0	22.2	23.7	24.8	25.0	24.7	23.9	22.7	21.5	20.8
	2%	DB	23.5	25.1	27.3	29.6	32.6	35.4	36.2	36.7	34.7	31.2	26.3	24.2
		MCWB	18.1	18.5	19.5	21.0	23.4	25.0	25.2	24.9	24.3	22.6	20.0	19.9
	5%	DB	21.7	23.0	25.7	28.3	31.5	33.9	34.9	35.3	33.3	29.5	24.5	22.3
		MCWB	17.7	18.3	18.7	20.6	23.4	24.9	25.3	25.3	24.3	21.7	18.9	18.9
	10%	DB	19.6	21.2	23.8	26.9	30.3	32.7	33.7	34.1	32.0	27.8	22.7	20.4
		MCWB	16.1	17.6	18.0	19.9	23.0	24.7	25.4	25.3	23.8	21.0	18.5	17.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.1	21.7	22.1	24.3	26.1	27.2	27.5	27.5	26.7	25.6	23.0	22.2
		MCDB	24.1	24.4	25.5	28.0	31.2	32.0	32.6	32.7	31.6	30.1	25.7	24.9
	2%	WB	19.9	20.6	21.1	23.0	25.1	26.4	26.9	26.9	26.0	24.6	21.8	20.9
		MCDB	22.4	23.6	24.7	26.9	29.9	31.7	32.3	32.5	31.1	28.6	24.2	23.3
	5%	WB	18.6	19.5	20.3	22.3	24.5	25.9	26.4	26.5	25.4	23.7	20.6	19.7
		MCDB	20.8	22.1	23.8	26.1	29.3	31.2	32.1	32.3	30.4	27.4	23.5	22.2
	10%	WB	16.9	18.0	19.4	21.5	23.8	25.5	26.0	26.0	24.8	22.6	19.1	17.9
		MCDB	19.0	20.6	22.8	25.1	28.6	30.8	31.5	31.7	29.5	25.8	22.0	19.9
Mean Daily Temperature Range	5% DB	MDBR	12.0	12.0	12.7	13.0	12.4	11.4	11.3	12.0	12.6	14.4	13.5	12.0
		MCDBR	13.1	13.0	13.3	13.1	12.8	12.9	12.7	13.6	13.9	14.8	14.7	13.1
		MCWBR	8.7	7.8	6.8	6.3	5.2	4.5	4.0	4.2	5.1	7.0	8.8	9.1
	5% WB	MCDBR	10.4	10.4	10.9	10.7	11.1	11.1	11.2	11.7	11.4	11.4	11.8	11.3
		MCWBR	8.0	7.3	6.7	6.1	5.1	4.6	4.2	4.3	4.7	6.2	8.0	8.8
Clear-Sky Solar Irradiance	taub	0.329	0.351	0.371	0.413	0.440	0.478	0.502	0.495	0.443	0.372	0.353	0.341	
	taud	2.527	2.465	2.406	2.291	2.261	2.197	2.157	2.176	2.294	2.464	2.469	2.492	
	Ebn at Noon	899	908	910	880	855	818	796	798	828	873	861	863	
	Edh at Noon	86	100	114	133	138	146	152	147	126	99	90	85	
All-Sky Solar Radiation	RadAvg	2.76	3.17	4.39	5.44	6.04	6.10	6.05	5.70	5.02	4.27	3.18	2.44	
	RadStd	0.25	0.54	0.36	0.42	0.40	0.51	0.45	0.52	0.46	0.59	0.32	0.28	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+177	N/A
(47) Regional (25 neighbors)	+0.31	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+118	+77

Nomenclature: See separate page